Benjamin Ching

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Work Experience

Data Engineer Co-op September 2026 – Present
Samsung R&D Vancouver, BC

- Reduced migration uncertainty across 38 recurring GCS-to-S3 pipelines by designing and testing a phased DataSync approach: single-file validation, manifest-based selective transfers, and batch-DAG shared-task concurrency testing.
- Designed a CI/CD automation proposal for an HQ-to-US BigQuery data-sync pipeline, targeting a 40% reduction in GitHub–Airflow configuration drift through automated Composer deployments.
- Engineered a Claude AI skill to automate user acceptance testing with documentation validation and rollback procedures with the help of a custom workflow bash script.
- Ideated a legacy-table detection algorithm that reconciles BigQuery inventory with active ingestion configurations, identifying 36 legacy tables across 141 assets; translating the approach into a reusable data-governance service that uses query history to assess downstream dependencies and prioritize retention, migration, or deprecation.

Systems Development and Data Analytics Intern May 2026 – August 2026
Wistron Remote

- Improved reliability of AI model data retrieval by replacing unstable JSON streams with PostgreSQL-backed queries and redesigning the Spring Boot Java backend.
- Implemented React and Tailwind CSS state-management mechanisms across 5+ internal user interfaces to improve consistency of dynamic frontend behavior.
- Prepared DIMM cold-plate datasets for analysis by cleaning, preprocessing, normalizing, filtering, and conducting EDA with Python, Pandas, and NumPy.
- Evaluated 8 decision-tree models against engineering requirements by selecting predictive features and incorporating physical estimates to target under 10% pressure change and 1.5°C thermal-resistance error.

Technical Projects

Chiron (Personal Project) | *Backend Engineering, MLOps, Data Pipeline Development, CI/CD* May 2026

- Built an end-to-end NBA fantasy analytics platform serving simulated performance data for 450 players with FastAPI, Pydantic, and PostgreSQL.
- Automated containerized build and deployment workflows with Docker and GitHub Actions, reducing manual releases from approximately 6 steps to 1 repeatable pipeline.
- Optimized data delivery by constructing PostgreSQL views to directly back live Power BI dashboards.

Nephos (nwHacks 2026) | *Cloud Data Engineering, Real-Time Monitoring, Serverless Architecture* Jan 2026

- Architected a real-time server-health monitoring and predictive-analytics dashboard, pitched directly to TELUS.
- Enabled near-real-time telemetry ingestion by building an AWS Lambda and EventBridge pipeline that streamed metrics into DynamoDB with sub-minute latency.
- Predicted approximately 80% of server-health issues by integrating Snowflake Cortex AI into the monitoring pipeline.

Helios (CalHacks 2025) | *Real-Time Systems, NLP/LLM Integration, Full-Stack Development* Oct 2025

- Engineered a real-time speech-to-graph pipeline that transformed live audio into visual knowledge maps using the Gemini LLM.
- Delivered transcript updates with under 50 ms latency by implementing WebSocket connections with VAPI.
- Supported knowledge maps with 5,000+ nodes by optimizing dynamic graph rendering with React Flow.

Technical Skills

Languages: Java, C/C++, Python, JavaScript/TypeScript, R, HTML/CSS, SQL
Frameworks: React, Node.js, Next.js, Express.js, FastAPI, REST APIs
Data/ML: PyTorch, scikit-learn, RAG, JupyterHub, Pandas, NumPy, Matplotlib, Power BI, NLP
Cloud Services: AWS (S3, Lambda, EventBridge, DataSync), Google Cloud (GCS, BigQuery, Cloud Composer)
Tools: Git/GitHub, Linux/Bash, Jira, Microsoft 365, gdb/valgrind, Claude Code/Codex

Education

University of British Columbia Nov. 2027 (3rd Year)
Bachelor of Science: Honours in Computer Science, Dean's List

Relevant Coursework: Object-Oriented Programming, Software Engineering, Data Structures and Algorithms, Hardware and Operating Systems, Relational Databases, Applied Machine Learning, Statistical Inference, Probability